

SELLER & INVENTORY MANAGEMENT SYSTEM

Database Documentation

1. Seller & Inventory Table Design

The Seller & Inventory Management System stores seller information, product details, and stock information. The database uses primary keys and foreign keys to maintain relationships between Seller, Product, and Inventory.

Seller Table

Column	Data Type	Constraint	Description
Seller_ID	INT	PRIMARY KEY	Unique seller identifier
Seller_Name	VARCHAR(100)	NOT NULL	Name of the seller
Email	VARCHAR(100)	UNIQUE, NOT NULL	Seller email address
Phone	VARCHAR(15)	UNIQUE, NOT NULL	Seller contact number
Address	VARCHAR(200)	NOT NULL	Seller location/address

Product Table

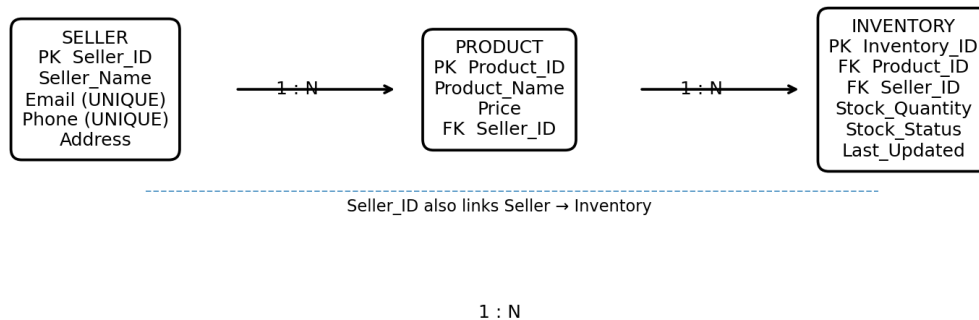
Column	Data Type	Constraint	Description
Product_ID	INT	PRIMARY KEY	Unique product identifier
Product_Name	VARCHAR(100)	NOT NULL	Name of product
Price	DECIMAL(10,2)	CHECK > 0	Product selling price
Seller_ID	INT	FOREIGN KEY	References Seller

Inventory Table

Column	Data Type	Constraint	Description
Inventory_ID	INT	PRIMARY KEY	Unique inventory record
Product_ID	INT	FOREIGN KEY	References Product
Seller_ID	INT	FOREIGN KEY	References Seller
Stock_Quantity	INT	CHECK >= 0	Available stock quantity
Stock_Status	VARCHAR(20)	NOT NULL	Available / Low Stock / Out of Stock
Last_Updated	DATE	—	Date when stock was last updated

2. Updated ER Diagram

The ER diagram shows the relationship between Seller, Product and Inventory. One seller can have many products, and inventory records are linked to both the seller and product.



3. Sample Seller & Inventory Data

Sample Seller Data

Seller_ID	Seller_Name	Email	Phone	Address
1	Subbu lakshmi	lakshmi@gmail.com	8190849810	Thoothukudi
2	manikandan	manikandan@gmail.com	6383430454	Tirunelveli
3	Sandhiya	sandhiya@gmail.com	9876543212	Coimbatore
4	Krishna	krish@gmail.com	9876543213	Salem

Sample Inventory Data

Inventory_ID	Product_ID	Seller_ID	Stock_Quantity	Stock_Status	Last_Updated
1	1001	1	25	Available	2026-09-20
2	1002	1	8	Low Stock	2026-09-20
3	1003	2	0	Out of Stock	2026-09-20
4	1004	3	50	Available	2026-09-20
5	1005	4	30	Available	2026-09-20

4. Inventory Analysis Report

The inventory analysis queries are used to monitor stock levels, identify low or out-of-stock products, calculate average stock, and identify the highest-stocked product.

Analysis	SQL Purpose	Expected Result from Sample Data
Available Stock	Stock_Quantity > 0	4 inventory records
Out of Stock	Stock_Quantity = 0	Product 1003
Low Stock	Stock_Quantity < 10	Product 1002
Highest Stock	ORDER BY Stock_Quantity DESC LIMIT 1	Product 1004 with 50 units
Average Stock	AVG(Stock_Quantity)	22.6 units

Key Observations

- Product 1004 has the highest stock quantity with 50 units.
- Product 1003 is currently out of stock.
- Product 1002 has low stock with 8 units before the update query.
- The stock update query changes Inventory_ID 2 from 8 units to 15 units.
- Foreign keys maintain the connection between sellers, products and inventory records.

5. Query Output Screenshots

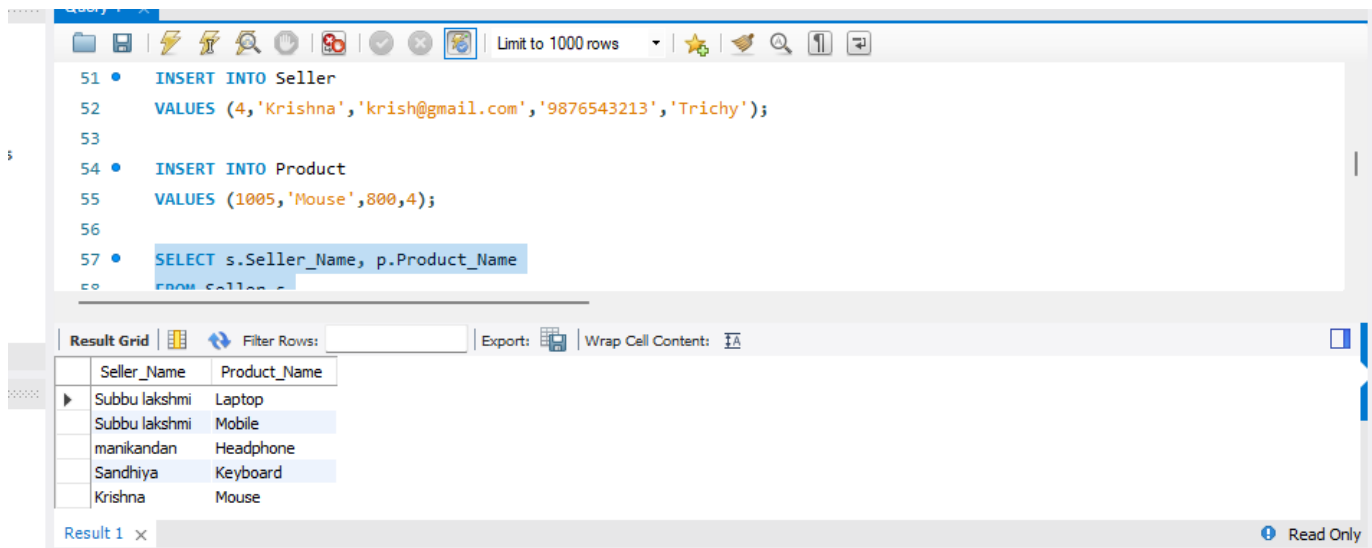
The following screenshots were supplied from the MySQL Workbench database work. They are included as evidence of table/query outputs and can be used in the project documentation.

6. SQL Script Contents

- Database and table creation
- Primary key, foreign key, NOT NULL, UNIQUE and CHECK constraints
- Seller, Product and Inventory sample INSERT queries
- Seller and Inventory CRUD operations
- Seller-wise product count
- Stock availability, low-stock and out-of-stock reports
- Highest-stocked product and average inventory quantity analysis

7. Conclusion

The Seller & Inventory Management System provides a structured way to store seller and product information and monitor stock levels. SQL constraints improve data integrity, while CRUD and analysis queries support routine inventory management and reporting.



The screenshot displays the MySQL Workbench interface. The top pane shows a SQL script with the following queries:

```
51 • INSERT INTO Seller
52   VALUES (4, 'Krishna', 'krish@gmail.com', '9876543213', 'Trichy');
53
54 • INSERT INTO Product
55   VALUES (1005, 'Mouse', 800, 4);
56
57 • SELECT s.Seller_Name, p.Product_Name
58   FROM Seller s
```

The bottom pane shows the 'Result Grid' with the following data:

Seller_Name	Product_Name
Subbu lakshmi	Laptop
Subbu lakshmi	Mobile
manikandan	Headphone
Sandhiya	Keyboard
Krishna	Mouse

The interface includes a toolbar at the top with icons for file operations, a 'Limit to 1000 rows' dropdown, and a 'Read Only' status indicator at the bottom right.

Query 1 x

Limit to 1000 rows

```
62 • SELECT s.Seller_Name,
63 COUNT(p.Product_ID) AS Product_Count
64 FROM Seller s
65 LEFT JOIN Product p
66 ON s.Seller_ID = p.Seller_ID
67 GROUP BY s.Seller_Name;
68
69 Update Seller Details
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Seller_Name	Product_Count
Subbu lakshmi	2
manikandan	1
Sandhiya	1
Krishna	1

Result 2 x Read Only

Query 2 x

Limit to 1000 rows

```
76
77 • SELECT *
78 FROM Inventory
79 WHERE Stock_Quantity = 0;
80
81 -- Products with Stock Less Than 10
82 • SELECT *
83 FROM Inventory
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

Inventory_ID	Product_ID	Seller_ID	Stock_Quantity	Stock_Status	Last_Updated
3	1003	2	0	Out of Stock	2026-09-20
NULL	NULL	NULL	NULL	NULL	NULL

Inventory 4 x Apply

Query 1 x

79 WHERE Stock_Quantity = 0;

80

81 • SELECT *

82 FROM Inventory

83 WHERE Stock_Quantity < 10;

84

85 -- Update Stock Quantity

86 • UPDATE Inventory

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	Inventory_ID	Product_ID	Seller_ID	Stock_Quantity	Stock_Status	Last_Updated
▶	2	1002	1	8	Low Stock	Low Stock 10
	3	1003	2	0	Out of Stock	2026-09-20
*	NULL	NULL	NULL	NULL	NULL	NULL

Inventory 5 x

Navigator:

- MANAGEMENT
 - Server Status
 - Client Connections
 - Users and Privileges
 - Status and System Variables
 - Data Export
 - Data Import/Restore
- INSTANCE
 - Startup / Shutdown
 - Server Logs
 - Online File
 - Administration Schemas
- Information
 - No object selected

Object Info Session

Query 1 x

80

81 SELECT *

82 FROM Inventory

83 WHERE Stock_Quantity < 10;

84

85 UPDATE Inventory

86 SET Stock_Quantity = 15

87 WHERE Inventory_ID = 2;

88

89 • DELETE FROM Inventory

90 WHERE Stock_Status = 'Out of Stock';

91

92 • SELECT s.Seller_Name, p.Product_Name

93 FROM Seller s

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

#	Time	Action	Message	Duration / Fetch
4	12:11:34	SELECT s.Seller_Name, p.Product_Name FROM Seller s JOIN Product p ON s.Seller_ID = p...	Error Code: 1046. No database selected Select the default DB to be used by double-clicking i...	0.000 sec
5	12:14:35	DELETE FROM Inventory WHERE Stock_Status = 'Discontinued'	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds t...	0.000 sec
6	12:17:10	DELETE FROM Inventory WHERE Stock_Status = 'Discontinued'	Error Code: 1046. No database selected Select the default DB to be used by double-clicking i...	0.000 sec
7	12:18:52	DELETE FROM Inventory WHERE Stock_Status = 'Discontinued'	Error Code: 1046. No database selected Select the default DB to be used by double-clicking i...	0.000 sec
8	12:20:39	DELETE FROM Inventory WHERE Stock_Status = 'Out of Stock'	Error Code: 1046. No database selected Select the default DB to be used by double-clicking i...	0.000 sec